WEST BENGAL

BEWARE OF THAT MAACHH-BHAAT

By Romita Datta

In Bengal, they say even the most exacting of palates can be satiated with a hearty meal of *maachh-bhaat,* or fish and rice. However, in a pincer attack of sorts, both face a threat of contamination by chemical poisoning. While arsenic in its inorganic form, occurring in soil and groundwater, is the blight that is poisoning varieties of paddy, mercury in water and the use of formalin are the villains infecting fish. Alarms raised in either case have failed to elicit an appropriate response.

It's known since the 1980s that large swathes of West Bengal fall in the naturally arsenic-contaminated Ganga-Brahmaputra-Meghna basin. Groundwater infected by arsenic and its consumption has been a major health concern for long. The entry of arsenic in the food chain through its presence in paddy grain has been established in research by scientists of the School of Environmental Studies (SOES) in Jadavpur University for the past several years. Dr Tarit Roychowdhury, senior faculty member of the school and water expert, who has studied arsenic contamination in soil and water and its effects on crops and vegetables, found out that eight rice-producing districts—North and South 24 Parganas, Nadia, Howrah, Hooghly, Bardhaman, Malda and Murshidabad—lie in the state's arsenic belt. Arsenic is listed as Group 1 carcinogen by the International Agency for Research on Cancer. It's linked to cancers of the skin, liver and kidney and can cause miscarriages and stillbirths. Crops contaminated with arsenic can, in the most alarming projection, slow-poison a mass of human population. Arsenic as an inorganic chemical is found in groundwater at a shallow depth of 40 metres. Most shallow tube-wells of Bengal are between 20 feet and 100 feet.

Crop contamination happens when arsenic-laden groundwater is used for irrigation. Thus, pre-monsoon paddy cultivated with groundwater runs a high risk of getting polluted with arsenic. According to agricultural research, rice paddy can absorb arsenic 10 times more than wheat and barley, because its roots are required to stand in flooded fields for a considerable period, resulting in maximum absorption of arsenic. While studying the progress of arsenic from root to shoot, Roychowdhury has observed that while its mobility is highest at the roots, other parts of the plant are also susceptible to the carcinogenic effects of the chemical. Studies have revealed that the edible rice grain has the least amount of arsenic accumulation in comparison to leaves and the shoot, but still contains 50-1000 ppb (parts per billion) of arsenic, whereas the prescribed limit of inorganic arsenic by the European Commission (2015) and Codex Alimentarius Commission (2014) is 100-200 ppb depending on different geo-environmental conditions.



MOREISH...BUT SAFE?
The beloved rice and fish curry of Bengalis could be laced with poison

This is not all. Paddy shoots, used as food for livestock, thus contain a high percentage of arsenic, putting at risk the animals and humans who consume the meat/ milk of cows and goats. According to Roychowdhury, a study on livestock, mainly cows, has revealed that cow milk, dung, urine and tail hair have traces of arsenic. While infected goat meat contains an arsenic concentration of 89-124 ppb, chicken meat contains 51 to 163 ppb. Worryingly, if arsenic present in cow urine/ dung gets mixed with the soil, it can recharge the arsenic table and start a vicious cycle of contamination, says Antara Das, senior research fellow, SOES.